

Influenza at the human-animal interface

Summary and risk assessment, from 13 June to 7 July 2026¹

- **New human cases²:** From 13 June to 7 July 2026, based on reporting date, detections of influenza A(H5) in one human, influenza A(H9N2) in two humans, and an influenza A(H3N2) variant ((H3N2)v) virus in one human were officially reported.
- **Circulation of influenza viruses with zoonotic potential in animals:** High pathogenicity avian influenza (HPAI) events in poultry and non-poultry animal species continue to be reported to the World Organisation for Animal Health (WOAH).³ The Food and Agriculture Organization of the United Nations (FAO) also provides a global update on avian influenza viruses with pandemic potential.⁴ Additionally, low pathogenicity avian influenza viruses as well as swine influenza viruses continue to circulate in animal populations.
- **Risk assessment⁵:** There have been no reports of sustained human-to-human transmission associated with the above-mentioned human infection events. Based on information available at the time of this risk assessment update, the overall public health risk from currently known influenza A viruses detected at the human-animal interface has not changed and- At present, these viruses are not thought to be capable of sustained human-to-human transmission, although this could change as they evolve. Although human infections with viruses of animal origin are infrequent, they are not unexpected at the human-animal interface.
- **IHR compliance⁶:** This includes any influenza A virus that has demonstrated the capacity to infect a human and its haemagglutinin (HA) gene (or protein) is not a mutated form of those, i.e. A(H1) or A(H3), circulating widely in the human population. Information from these notifications is critical to inform risk assessments for influenza at the human-animal interface.

Avian influenza viruses in humans

A(H5), Bangladesh

On 15 June 2026, Bangladesh notified WHO of one laboratory-confirmed human case of avian influenza A(H5) infection in Bangladesh in a child from Sylhet Division. The case was detected notified through the National Influenza Surveillance, Bangladesh (NISB) platform as an influenza like-illness (ILI) case.

¹ This summary and assessment covers information confirmed during this period and may include information received outside of this period.

² For epidemiological and virological features of human infections with animal influenza viruses not reported in this assessment, see the reports on human cases of influenza at the human-animal interface published in the Weekly Epidemiological Record [here](#).

³ World Organisation for Animal Health (WOAH). Avian influenza. Global situation. Available at: <https://www.woah.org/en/disease/avian-influenza/#ui-id-2>.

⁴ Food and Agriculture Organization of the United Nations (FAO). Global Avian Influenza Viruses with Zoonotic Potential situation update. Available at: <https://www.fao.org/animal-health/situation-updates/global-aiv-with-zoonotic-potential>.

⁵ World Health Organization (2012). Rapid risk assessment of acute public health events. World Health Organization. Available at: <https://iris.who.int/handle/10665/70810>.

⁶ World Health Organization. Case definitions for the four diseases requiring notification in all circumstances under the International Health Regulations (2005). Available at: [https://www.who.int/publications/m/item/case-definitions-for-the-four-diseases-requiring-notification-to-who-in-all-circumstances-under-the-ihl-\(2005\)](https://www.who.int/publications/m/item/case-definitions-for-the-four-diseases-requiring-notification-to-who-in-all-circumstances-under-the-ihl-(2005)).

The patient developed respiratory symptoms on 17 May 2026, received outpatient healthcare on 20 May. A clinical sample was collected that day and was received by the Institute of Epidemiology, Disease Control and Research (IEDCR) on 4 June as part of routine surveillance. The sample tested positive for influenza A(H5) virus by real-time reverse transcription polymerase chain reaction (RT-PCR) on 11 June.

The patient is now in good health and reported no travel history and no history of exposure to poultry. However, poultry deaths were reported in the area surrounding the patient's residence. The outbreak investigation team identified and followed close and possible contacts. Samples from some of the close contacts as well as animal and environmental samples were collected for testing for influenza. All contacts remained asymptomatic and all samples tested negative for influenza.

This is the third laboratory-confirmed human case of avian influenza A(H5) reported in Bangladesh in 2026, and the 15th human case of avian influenza A(H5) reported to WHO from Bangladesh since 2008, including two fatal cases, one reported in 2013 and one in 2026.

Risk assessment for avian influenza A(H5) viruses:

1. What is the current global public health risk of additional human cases of infection with avian influenza A(H5) viruses?

Most human infections so far have been reported in people exposed to A(H5) viruses, for example, through contact with infected poultry or contaminated environments, including live poultry markets, and occasionally infected mammals and contaminated environments. As long as the viruses continue to be detected in animals and related environments humans are exposed to, further human cases associated with such exposures are expected but remain unusual. The impact for public health if additional sporadic cases are detected is minimal. The current overall global public health risk is low.

2. What is the likelihood of sustained human-to-human transmission of avian influenza A(H5) viruses related to the events above?

No sustained human-to-human transmission has been identified associated with the recent reported human infections with avian influenza A(H5) viruses. There has been no reported human-to-human transmission of A(H5N1) viruses since 2007, although there may be gaps in investigations. In 2007 and the years prior, small clusters of A(H5) virus infections in humans were reported, including some involving health care workers, where limited human-to-human transmission could not be excluded; however, sustained human-to-human transmission was not reported. Current evidence suggests that influenza A(H5) viruses related to these events did not acquire the ability to efficiently transmit between people.

3. What is the likelihood of international spread of avian influenza A(H5) viruses by travellers?

Should infected individuals from affected areas travel internationally, their infection may be detected in another country during travel or after arrival. If this were to occur, further community-level spread is considered unlikely as current evidence suggests these viruses have not acquired the ability to transmit easily among humans.

A(H9N2), China

Between 12 and 23 June 2026, two laboratory-confirmed cases of A(H9N2) virus infection were detected in China. Both cases had mild illness and were hospitalized in isolation wards at the time of reporting.

Age and sex	Area	Symptom onset
< 5-year-old male	Hong Kong Special Administrative Region (SAR)	9 June 2026
< 5-year-old female	Guangdong	12 June 2026

Both cases had exposure to local live bird markets. Samples from environments associated with the likely area of exposure of the cases tested positive for A(H9) viruses. No further cases were detected among contacts of these cases.

Risk assessment for avian influenza A(H9N2):

1. What is the global public health risk of additional human cases of infection with avian influenza A(H9N2) viruses?

Most human cases follow exposure to the A(H9N2) virus through contact with infected poultry or contaminated environments. Most human infections of A(H9N2) to date have resulted in mild clinical illness. Since the virus is endemic in poultry in multiple countries in Africa and Asia, additional human cases associated with exposure to infected poultry or contaminated environments are expected but remain unusual. The impact to public health if additional sporadic cases are detected is minimal. The overall global public health risk is low.

2. What is the likelihood of sustained human-to-human transmission of avian influenza A(H9N2) viruses related to these events?

At the present time, no sustained human-to-human transmission has been identified associated with the recently reported human infections with A(H9N2) viruses. Current evidence suggests that A(H9N2) viruses from these cases did not acquire the ability of sustained transmission among humans.

3. What is the likelihood of international spread of avian influenza A(H9N2) virus by travellers?

Should infected individuals from affected areas travel internationally, their infection may be detected in another country during travel or after arrival. If this were to occur, further community level spread is considered unlikely as current evidence suggests the A(H9N2) virus subtype has not acquired the ability to transmit easily among humans.

Swine influenza viruses in humans

Influenza A(H3N2)v, Brazil

On 25 June 2026, Brazil notified PAHO/WHO of a laboratory-confirmed human infection with an influenza A(H3N2)v virus detected in a child in Santa Catarina state. The patient had symptom onset on 12 June 2026 and due to worsening respiratory symptoms, healthcare was sought on 16 June. The patient was referred for hospital admission with a diagnosis of Severe Acute Respiratory Infection (SARI). Upon admission, an antigen test confirmed influenza A and the patient was placed in a private respiratory isolation room and antiviral treatment was initiated. The patient was discharged on 19 June.

A nasopharyngeal swab sample was collected on 16 June and sent to the State public health laboratory for real-time RT-PCR. On 18 June, a swine-origin influenza H3 variant was suspected, and the sample was sent to the Laboratory of Respiratory Viruses, Exanthems, Enteroviruses, and Viral Emergencies (LVRE) at the Oswaldo Cruz Institute (Fiocruz/Rio de Janeiro) on 19 June. Analyses confirmed the presence of an influenza A(H3N2)v virus via molecular testing and genomic sequencing.

An investigation by the state and municipality epidemiological surveillance team found that all contacts were asymptomatic before, during and after the child's illness. The child's grandfather

worked at a swine nursery housing approximately 5,000 animals, though he noted that sanitary barriers were in place. The child frequently visited the grandfather's home and had contact with him several days a week.

This is the first human A(H3N2)v infection detected in the Brazil in 2026 and the first case reported in the state of Santa Catarina.

Risk assessment for swine influenza viruses:

1. What is the public health risk of additional human cases of infection with swine influenza viruses?

Swine influenza viruses circulate in swine populations in many regions of the world. Depending on geographic location, the genetic characteristics of these viruses differ. Most human cases are exposed to swine influenza viruses through contact with infected animals or contaminated environments. Human infection tends to result in mild clinical illness in most cases. Since these viruses continue to be detected in swine populations, further human cases are expected. The impact to public health if additional sporadic cases are detected is minimal. The overall risk of additional sporadic human cases is low.

2. What is the likelihood of sustained human-to-human transmission of swine influenza viruses?

No sustained human-to-human transmission was identified associated with the event described above. Current evidence suggests that contemporary swine influenza viruses have not acquired the ability of sustained transmission among humans.

3. What is the likelihood of international spread of swine influenza viruses by travellers?

Should infected individuals from affected areas travel internationally, their infection may be detected in another country during travel or after arrival. If this were to occur, further community level spread is considered unlikely as current evidence suggests that these viruses have not acquired the ability to transmit easily among humans.

Overall risk management recommendations:

Surveillance and investigations

- Due to the constantly evolving nature of influenza viruses, WHO continues to stress the importance of global strategic surveillance in animals and humans to detect virologic, epidemiologic and clinical changes associated with circulating influenza viruses that may affect human (or animal) health. Continued vigilance is needed within affected and neighbouring areas to detect infections in animals and humans. Close collaboration with the animal health and environment sectors is essential to understand the extent of the risk of human exposure and to prevent and control the spread of animal influenza. WHO has published guidance on [surveillance for human infections with avian influenza A\(H5\) viruses](#).
- As the extent of influenza virus circulation in animals is not clear, epidemiologic and virologic surveillance and the follow-up of suspected human cases should continue systematically. [Guidance on investigation of non-seasonal influenza and other emerging acute respiratory diseases](#) has been published on the WHO website.
- Countries should increase avian influenza surveillance in domestic and wild birds, enhance surveillance for early detection in cattle populations in countries where HPAI is known to be circulating, include HPAI as a differential diagnosis in non-avian species, including cattle and other livestock populations, with high risk of exposure to HPAI viruses; monitor and investigate cases in non-avian species, including livestock, report cases of HPAI in all animal species, including unusual hosts, to WOA and other international organizations, share genetic sequences of avian influenza viruses in publicly available databases, implement preventive and

early response measures to break the HPAI transmission cycle among animals through movement restrictions of infected livestock holdings and strict biosecurity measures in all holdings, employ good production and hygiene practices when handling animal products, and protect persons in contact with suspected/infected animals.⁷ More guidance can be found from [WOAH](#) and [FAO](#).

- When there has been human exposure to a known outbreak of an influenza A virus in domestic poultry, wild birds or other animals – or when there has been an identified human case of infection with such a virus – enhanced surveillance in potentially exposed human populations becomes necessary. Enhanced surveillance should consider the health care seeking behaviour of the population, and could include a range of active and passive health care and/or community-based approaches, including: enhanced surveillance in local influenza-like illness (ILI)/SARI systems, active screening in hospitals and of groups that may be at higher occupational risk of exposure, and inclusion of other sources such as traditional healers, private practitioners and private diagnostic laboratories.
- Vigilance for the emergence of novel influenza viruses with pandemic potential should be maintained at all times including during a non-influenza emergency. In the context of the co-circulation of SARS-CoV-2 and influenza viruses, WHO has updated and published [practical guidance for integrated surveillance](#).

Notifying WHO

- All human infections caused by a new subtype of influenza virus are notifiable under the International Health Regulations (IHR, 2005).^{8,9} State Parties to the IHR (2005) are required to immediately notify WHO of any laboratory-confirmed¹⁰ case of a recent human infection caused by an influenza A virus with the potential to cause a pandemic¹¹. Evidence of illness is not required for this report. Evidence of illness is not required for this report.
- WHO published the case definition for human infections with avian influenza A(H5) virus requiring notification under IHR (2005): <https://www.who.int/teams/global-influenza-programme/avian-influenza/case-definitions>.

Virus sharing and risk assessment

- It is critical that these influenza viruses from animals or from humans are fully characterized in appropriate animal or human health influenza reference laboratories. Under WHO's Pandemic Influenza Preparedness (PIP) Framework, Member States are expected to share influenza viruses with pandemic potential on a **timely basis**¹² with a WHO Collaborating Centre for influenza of GISRS. The viruses are used by the public health laboratories to assess the risk of pandemic influenza and to develop candidate vaccine viruses.

⁷ World Organisation for Animal Health. Statement on High Pathogenicity Avian Influenza in Cattle, 6 December 2024 (<https://www.woah.org/en/high-pathogenicity-avian-influenza-hpai-in-cattle/>).

⁸ World Health Organization. International Health Regulations (2005), as amended through resolutions WHA67.13 (2014), WHA75.12 (2022), and WHA77.17 (2024) (https://apps.who.int/gb/bd/pdf_files/IHR_2014-2022-2024-en.pdf).

⁹ World Health Organization. Case definitions for the four diseases requiring notification in all circumstances under the International Health Regulations (2005) (<https://www.who.int/publications/m/item/case-definitions-for-the-four-diseases-requiring-notification-to-who-in-all-circumstances-under-the-ihp-2005>).

¹⁰ World Health Organization. Manual for the laboratory diagnosis and virological surveillance of influenza (2011) (<https://apps.who.int/iris/handle/10665/44518>).

¹¹ World Health Organization. Pandemic influenza preparedness framework for the sharing of influenza viruses and access to vaccines and other benefits, 2nd edition (<https://iris.who.int/handle/10665/341850>).

¹² World Health Organization. Operational guidance on sharing influenza viruses with human pandemic potential (IVPP) under the Pandemic Influenza Preparedness (PIP) Framework (2017) (<https://apps.who.int/iris/handle/10665/259402>).

- The Tool for Influenza Pandemic Risk Assessment (TIPRA) provides an in-depth assessment of risk associated with some zoonotic influenza viruses – notably the likelihood of the virus gaining human-to-human transmissibility, and the impact should the virus gain such transmissibility. TIPRA maps relative risk amongst viruses assessed using multiple risk elements. The results of TIPRA complement those of the risk assessment provided here, and those of prior TIPRA risk assessments are published at [http://www.who.int/teams/global-influenza-programme/avian-influenza/tool-for-influenza-pandemic-risk-assessment-\(tipra\)](http://www.who.int/teams/global-influenza-programme/avian-influenza/tool-for-influenza-pandemic-risk-assessment-(tipra)).

Risk reduction

- Given the observed extent and frequency of avian influenza in poultry, wild birds and some wild and domestic mammals, the public should avoid contact with animals that are sick or dead from unknown causes, including wild animals, and should report dead birds and mammals or request their removal by contacting local wildlife or veterinary authorities.
- Eggs, poultry meat and other poultry food products should be properly cooked and properly handled during food preparation. Due to the potential health risks to consumers, raw milk should be avoided. WHO advises consuming pasteurized milk. If pasteurized milk isn't available, heating raw milk until it boils makes it safer for consumption.
- WHO has published [practical interim guidance to reduce the risk of infection in people exposed to avian influenza viruses](#).

Trade and travellers

- WHO advises that travellers to countries with known outbreaks of animal influenza should avoid farms, contact with animals in live animal markets, entering areas where animals may be slaughtered, or contact with any surfaces that appear to be contaminated with animal excreta. Travelers should also wash their hands often with soap and water. All individuals should follow good food safety and hygiene practices.
- WHO does not advise special traveller screening at points of entry or restrictions with regards to the current situation of influenza viruses at the human-animal interface. For recommendations on safe trade in animals and related products from countries affected by these influenza viruses, refer to [WOAH](#) guidance.

Links:

WHO Human-Animal Interface web page

<https://www.who.int/teams/global-influenza-programme/avian-influenza>

WHO Influenza (Avian and other zoonotic) fact sheet

[https://www.who.int/news-room/fact-sheets/detail/influenza-\(avian-and-other-zoonotic\)](https://www.who.int/news-room/fact-sheets/detail/influenza-(avian-and-other-zoonotic))

WHO Protocol to investigate non-seasonal influenza and other emerging acute respiratory diseases

<https://www.who.int/publications/i/item/WHO-WHE-IHM-GIP-2018.2>

WHO Public health resource pack for countries experiencing outbreaks of influenza in animals:

<https://www.who.int/publications/i/item/9789240076884>

Cumulative Number of Confirmed Human Cases of Avian Influenza A(H5N1) Reported to WHO

<https://www.who.int/teams/global-influenza-programme/avian-influenza/avian-a-h5n1-virus>

Avian Influenza A(H7N9) Information

[https://www.who.int/teams/global-influenza-programme/avian-influenza/avian-influenza-a-\(h7n9\)-virus](https://www.who.int/teams/global-influenza-programme/avian-influenza/avian-influenza-a-(h7n9)-virus)

World Organisation of Animal Health (WOAH) web page: Avian Influenza

<https://www.woah.org/en/home/>

Food and Agriculture Organization of the United Nations (FAO) webpage: Avian Influenza

<https://www.fao.org/animal-health/avian-flu-qa/en/>

WOAH/FAO Network of Expertise on Animal Influenza (OFFLU)

<http://www.offlu.org/>